Inhibition of angiotensin convertin enzyme (ACE) activity by the anthocyanins delphinidin- and cyanidin-3-O-sambubiosides from Hibiscus sabdariffa.

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ETHNOPHARMACOLOGICAL RELEVANCE: The beverages of Hibiscus sabdariffa calyces are widely used in Mexico as diuretic, for treating gastrointestinal disorders, liver diseases, fever, hypercholesterolemia and hypertension. Different works have demonstrated that Hibiscus sabdariffa extracts reduce blood pressure in humans, and recently, we demonstrated that this effect is due to angiotensin converting enzyme (ACE) inhibitor activity. AIM OF THE STUDY: The aim of the current study was to isolate and characterizer the constituents responsible of the ACE activity of the aqueous extract of Hibiscus sabdariffa. MATERIALS AND METHODS: Bioassay-guided fractionation of the aqueous extract of dried calyces of Hibiscus sabdariffa using preparative reversed-phase HPLC, and the in vitro ACE Inhibition assay, as biological monitor model, were used for the isolation. The isolated compounds were characterized by spectroscopic methods. RESULTS: The anthocyanins delphinidin-3-O-sambubioside (1) and cyanidin-3-O-sambubioside (2) were isolated by bioassay-guided purification. These compounds showed IC(50) values (84.5 and 68.4 microg/mL, respectively), which are similar to those obtained by related flavonoid glycosides. Kinetic determinations suggested that these compounds inhibit the enzyme activity by competing with the substrate for the active site. CONCLUSIONS: The competitive ACE inhibitor activity of the anthocyanins 1 and 2 is reported for the first time. This activity is in good agreement with the folk medicinal use of Hibiscus sabdariffa calyces as antihypertensive.